WORKED SOLUTION

STEP II 2008, Question 5

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q5

The key idea

Both integrals yield to spotting a derivative. In the first, the numerator $\sin 2x = 2\sin x \cos x$ is exactly the derivative of the denominator $1 + \sin^2 x$, so the integral is a logarithm on sight. In the second, the lone $\sin x$ on top suggests $c = \cos x$, after writing $1 + \sin^2 x = 2 - \cos^2 x$, which leaves a standard partial-fractions integral. Comparing the two answers then turns into comparing two numbers involving $\sqrt{2}$, and the inequalities in the final part are built to settle exactly that.

Solution

The two integrals

Call the integrals I_1 (numerator $\sin 2x$) and I_2 (numerator $\sin x$).

For I_1, since $\dfrac{\mathrm{d}}{\mathrm{d}x}(1 + \sin^2 x) = 2\sin x \cos x = \sin 2x$, the integrand has the form $f'(x)/f(x)$ with $f(x) = 1 + \sin^2 x > 0$. Hence

$$I_1 = \Big[\ln(1 + \sin^2 x)\Big]_0^{\pi/2} = \ln 2 - \ln 1 = \ln 2.$$

For I_2, write $1 + \sin^2 x = 2 - \cos^2 x$ and substitute $c = \cos x$, so $\mathrm{d}c = -\sin x\,\mathrm{d}x$; as x runs from 0 to $\frac{1}{2}\pi$, c runs from 1 down to 0. The minus sign reverses the limits:

$$I_2 = \int_0^1 \frac{\mathrm{d}c}{2 - c^2}.$$

Using partial fractions,

$$\frac{1}{2 - c^2} = \frac{1}{(\sqrt{2} - c)(\sqrt{2} + c)} = \frac{1}{2\sqrt{2}}\left(\frac{1}{\sqrt{2} - c} + \frac{1}{\sqrt{2} + c}\right),$$

and both denominators stay positive for $0 \le c \le 1$, so

$$I_2 = \frac{1}{2\sqrt{2}}\Big[\ln(\sqrt{2}+c) - \ln(\sqrt{2}-c)\Big]_0^1 = \frac{1}{2\sqrt{2}}\ln\frac{\sqrt{2}+1}{\sqrt{2}-1}.$$

This tidies up nicely: $(\sqrt{2}+1)(\sqrt{2}-1) = 1$, so $\dfrac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2}+1)^2$, and therefore

$$I_2 = \frac{1}{2\sqrt{2}} \cdot 2\ln(1+\sqrt{2}) = \frac{\ln(1+\sqrt{2})}{\sqrt{2}}.$$

The inequalities

By the binomial theorem, using $(\sqrt{2})^2 = 2$, $(\sqrt{2})^3 = 2\sqrt{2}$, $(\sqrt{2})^4 = 4$ and $(\sqrt{2})^5 = 4\sqrt{2}$,

$$\begin{aligned}(1+\sqrt{2})^5 &= 1 + 5\sqrt{2} + 10\cdot 2 + 10\cdot 2\sqrt{2} + 5\cdot 4 + 4\sqrt{2}\\ &= 41 + 29\sqrt{2}.\end{aligned}$$

Now $\sqrt{2} < 2$ (as $2 < 4$), so $29\sqrt{2} < 58$ and $(1+\sqrt{2})^5 < 41 + 58 = 99$, as required.

Next, $1.4^2 = 1.96 < 2$, and since both 1.4 and $\sqrt{2}$ are positive, squaring preserves order, so $\sqrt{2} > 1.4$.

Ask yourself How can $(1+\sqrt{2})^5 < 99$ and $\sqrt{2} > 1.4$ possibly tell us anything about $2^{\sqrt{2}}$?

Write $1.4 = \frac{7}{5}$. Then $2^{7/5}$ is the fifth root of $2^7 = 128$, and the fifth power of $1+\sqrt{2}$ is what we have just bounded. Precisely: since $t \mapsto 2^t$ is increasing and $\sqrt{2} > \frac{7}{5}$,

$$2^{\sqrt{2}} > 2^{7/5}.$$

Also $(1+\sqrt{2})^5 < 99 < 128 = (2^{7/5})^5$, and since $y \mapsto y^5$ is increasing for $y > 0$, taking positive fifth roots gives $1+\sqrt{2} < 2^{7/5}$. Combining,

$$2^{\sqrt{2}} > 2^{7/5} > 1 + \sqrt{2}.$$

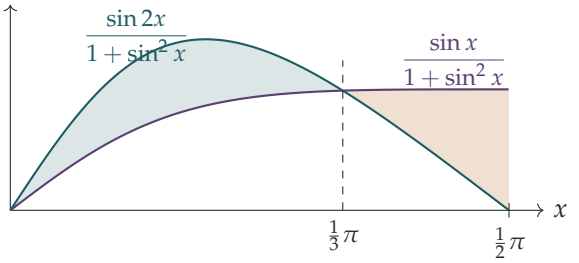
Which integral is greater?

The function ln is increasing, so the last inequality gives $\ln(2^{\sqrt{2}}) > \ln(1+\sqrt{2})$, that is, $\sqrt{2}\ln 2 > \ln(1+\sqrt{2})$. Dividing by $\sqrt{2} > 0$,

$$I_1 = \ln 2 > \frac{\ln(1+\sqrt{2})}{\sqrt{2}} = I_2.$$

So the integral with $\sin 2x$ in the numerator is the greater.

The comparison is not something we could have read off the integrands directly. Since $\sin 2x - \sin x = \sin x\,(2\cos x - 1)$, the first integrand is the larger for $0 < x < \frac{1}{3}\pi$ and the smaller for $\frac{1}{3}\pi < x < \frac{1}{2}\pi$:



The question is whether the teal region outweighs the amber one, and it does, but only by about 0.07. That is why some careful arithmetic with $\sqrt{2}$ was needed.

Checks

- Simpson's rule on the original integrands gives $I_1 \approx 0.693147$ and $I_2 \approx 0.623225$, matching $\ln 2$ and $\ln(1+\sqrt{2})/\sqrt{2}$ to ten decimal places.
- $(1+\sqrt{2})^5 \approx 82.01$ and $41 + 29\sqrt{2} \approx 82.01$; well below 99.
- $2^{\sqrt{2}} \approx 2.665$, $2^{1.4} \approx 2.639$ and $1+\sqrt{2} \approx 2.414$, in the claimed order.
- Sanity check on I_2: the integrand lies between $\frac{1}{2}\sin x$ and $\sin x$, so $\frac{1}{2} < I_2 < 1$, as 0.62 does.

Common slips

- Forgetting that $\mathrm{d}c = -\sin x\,\mathrm{d}x$ flips the limits, and ending with a negative value for a positive integrand.
- Leaving I_2 as $\frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2}+1}{\sqrt{2}-1}$: without spotting that this is $(\sqrt{2}+1)^2$, the link to $1+\sqrt{2}$ in the final part is hidden.
- Asserting $2^{\sqrt{2}} > 1+\sqrt{2}$ from a calculator. The question wants the chain through $2^{7/5}$, with the monotonicity of 2^t and y^5 stated.

Beyond STEP

What does $2^{\sqrt{2}}$ even mean? At university it is *defined* as $e^{\sqrt{2}\ln 2}$, and the fact we used, that 2^t is increasing in t for irrational as well as rational t, follows from properties of the exponential function that a first analysis course proves carefully. (A famous theorem of Gelfond and Schneider shows that $2^{\sqrt{2}}$ is in fact transcendental.) For the careful construction of e^x and $\ln x$, see Abbott, *Understanding Analysis*.

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